DG INTERN HUB

SOC ANALYST INTERNSHIP

WEEK 2 — TRYHACKME ROOM WRITEUPS

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Domain: Cybersecurity / SOC Analyst

Platform: TryHackMe

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ROOM 6: HTTP IN DETAIL

Learning Objective

The objective of this room was to understand how HTTP communication works, learn about HTTP methods and status codes, and understand how requests and responses are useful in web security investigations.

1. Introduction

In this room, I learned how web browsers and web servers communicate using HTTP. Whenever we open a website, submit a form, or access a web page, HTTP requests and responses help transfer information between the client and server.

I also learned how HTTPS protects communication by using encryption through TLS.

2. What Is HTTP and HTTPS?

HTTP (HyperText Transfer Protocol): A protocol used for communication between web clients and web servers.

HTTPS (HyperText Transfer Protocol Secure): The secure version of HTTP, which uses TLS to encrypt communication and help protect data in transit.

HTTPS helps protect sensitive information from being read or modified by unauthorised parties during transmission.

3. HTTP Requests and Responses

An HTTP request is sent by a client, such as a browser, to a web server. The server processes the request and returns an HTTP response.

An HTTP request can contain:

- Request method
- URL or path
- HTTP version
- Headers
- Optional body

An HTTP response can contain:

- HTTP version
- Status code
- Response headers
- Optional response body

4. HTTP Methods

- **GET:** Requests data from a server.
- **POST:** Sends data to a server, often to create a resource or submit a form.
- **PUT:** Creates or replaces a resource at a specified location, depending on the server's implementation.
- **DELETE:** Requests the removal of a resource.

These methods help identify what action a client is requesting from a web server.

5. HTTP Status Codes

HTTP status codes indicate the result of a request.

- **200 OK:** The request was successful.
- **201 Created:** The request successfully created a resource.
- **301 Moved Permanently:** The requested resource has a permanent new location.
- **400 Bad Request:** The server could not process the request because of a client-side error.
- **401 Unauthorized:** Authentication is required or has failed.
- **403 Forbidden:** The server refuses to allow access.
- **404 Not Found:** The requested resource could not be found.
- **500 Internal Server Error:** The server encountered an unexpected error.
- **503 Service Unavailable:** The server is temporarily unable to handle the request.

6. HTTP Headers

Headers provide additional information about requests and responses.

- **Host:** Identifies the requested host.

- **User-Agent:** Provides information about the client software.
- **Content-Type:** Indicates the media type of the message body.
- **Content-Length:** Specifies the size of the message body in bytes.
- **Set-Cookie:** Allows a server to send a cookie to the client.

Headers help clients and servers exchange information about the request, response, and data being transferred.

7. Cookies

Cookies are small pieces of data that websites store in a user's browser. They can be used to maintain sessions, remember preferences, and support authentication.

Cookies are important during security investigations because stolen or misused session cookies may allow an attacker to impersonate a user.

8. Making HTTP Requests

During the practical exercises, I explored how different HTTP methods and parameters affect requests.

I learned that GET requests can include URL parameters, while POST and PUT requests commonly send data in the request body. The exact behaviour depends on how the server processes the request.

I also learned to inspect the server's response to understand whether a request succeeded or failed.

9. SOC Analyst Perspective

HTTP logs and request details can help SOC analysts investigate suspicious web activity. Analysts may look for unusual URLs, unexpected request methods, repeated failed requests, suspicious user agents, and unusual response codes.

For example, repeated requests to non-existent pages may indicate scanning or attempted discovery. A large number of failed authentication attempts may require further investigation.

10. What I Learned

- I understood the difference between HTTP and HTTPS.
- I learned how HTTP requests and responses work.
- I learned about GET, POST, PUT, and DELETE methods.
- I understood common HTTP status codes.
- I learned the purpose of important HTTP headers and cookies.
- I understood how web request information can help investigate suspicious activity.

11. Conclusion

This room helped me understand how clients and web servers communicate using HTTP. I learned that request methods,

headers, status codes, and cookies provide useful information for troubleshooting and security investigations.